

CSMFAB000000000103 — Shipping Container Safe Pod Redesign

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMFAB000000000103 — Shipping Container Safe Pod Redesign V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Primary structural material	Corten steel ($\rho=1.3\times10^{-7}\ \Omega\cdot\text{m}$, 2300 kg per TEU)	BFRP/Elium® ($\rho>10^{10}\ \Omega\cdot\text{m}$, 680 kg per TEU)	Steel hull couples GIC $\square$ Joule heating; BFRP eliminates path; 1620 kg mass saving	Arkema Elium® TDS 2025; Composites Part A 2024
Composite matrix recyclability	Thermoset epoxy: landfill EOL	Elium® thermoplastic: solvolysis 350°C, 95% fiber recovery	Phoenix Protocol mandate: thermoset has no viable recovery path	Arkema Elium® TDS 2025
Corner fittings	Steel ISO 1161 (conductive)	Flash-sintered 3Y-TZP ZrO ₂ ceramic (1200°C, $\rho>10^{12}\ \Omega\cdot\text{m}$, >1300 MPa)	Steel fittings stackable conductor loops in container city configuration	Johnson et al. 2024; ZrO ₂ 3Y-TZP data 2025
Aerogel specification	Silica aerogel $\lambda=0.015\ \text{W/m}\cdot\text{K}$, max 600°C	Polyimide-silica hybrid $\lambda=0.010\ \text{W/m}\cdot\text{K}$, max 650°C	33% lower conductivity; 50°C higher ceiling for fire scenario	Aspen Aerogels Pyrogel XT-E 2025 datasheet

Parameter	V1.0	V2.0	Reason	Primary Source
EMI shielding	Carbon-doped polymer (SE ~40 dB)	Ti ₃ C ₂ T _x MXene spray (SE 92 dB)	52 dB improvement; spray-applied on existing BFRP hull face	ACS Appl. Mater. Interfaces 17/8 (2025)
Post-event coordination	GPS/cellular dependent	LoRa 256-node mesh, GPS-free, 5 km per hop	GPS/cellular destroyed in Carrington scenario — primary use case	LoRa Alliance v1.1; ITU-R WP 4C 2024
Seismic stacking standard	ACI 440.1R-15	ACI 440.11-25 Seismic Type S provisions	Code superseded; seismic Type S required for SDC D/E/F multi-story stacking	ACI 440.11-25 January 2025

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Primary structural material	Steel hull couples GIC → Joule heating; BFRP eliminates path; 1620 kg mass saving
Composite matrix recyclability	Phoenix Protocol mandate: thermoset has no viable recovery path
Corner fittings	Steel fittings stackable conductor loops in container city configuration
Aerogel specification	33% lower conductivity; 50°C higher ceiling for fire scenario
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Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward un-**

changed from V1.0. Only the specific parameters listed above were revised.

*End of Versioning Delta | CSMFAB000000000103 — Shipping Container Safe Pod Redesign
V1.0 archived; V2.0 is the active specification as of June 2026.*